

FESTO Distributing station: Automatic Check of Process Implementability and Gap Identification

In our ongoing projects we develop an LLM-driven manufacturability analysis tool underpinned by a multi-layered ontology in GraphDB that models the product's features, the structural and functional properties of the production line, the available machinery, and the automation code. Festo MPS Distribution station could be taken as use case.

The system should let the user ask: *“Is this product or production process manufacturable on this line?”*—the LLM will reason over the ontological layers to detect missing capabilities, misaligned parameters, or absent control logic, then propose how to fill the gaps through new ontology links, equipment or tool additions, or generated IEC 61499/PLC code patches.

Optionally, the reasoning and proposed fixes can be validated in a 3D simulation environment such as the Open Industry Project or another web-based twin of a Festo MPS cell connected via OPC UA to a soft PLC.

We ask you to suggest ideas enhancing this approach on the (super-simplified) example of “FESTO Distributing Station”.

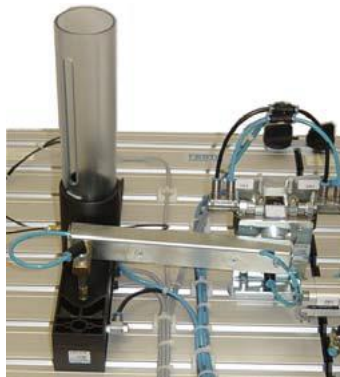


Figure 1. FESTO Distributing Station.

We developed an (incomplete but functional) program to control this station using the skills approach using IEC 61499 and EcoStruxure Automation Expert tool. Skills are unified stateful services “embedded” to or associated with mechatronic components.

Organising program using skills serves several purposes:

1. Each operation has the same exposed standardized interface. A skill can be invoked programmatically in IEC 61499 application, or via its faceplate, or via OPC UA. This helps to improve interoperability.
2. Overall structure of the system can be better understood by discovering relation between skills. These relations can be represented as a knowledge graph. In this exercise we will focus on this design time use of skills.

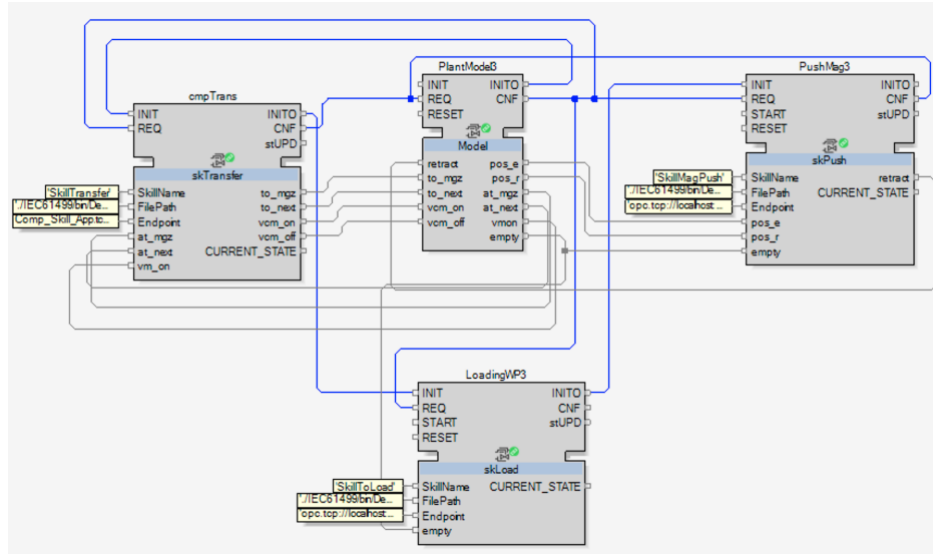


Figure 2. Control application (semi-automatic) of the FESTO Distributing station.

When running one can open the simulation Canvas and initiate the skills using their HMI faceplates.

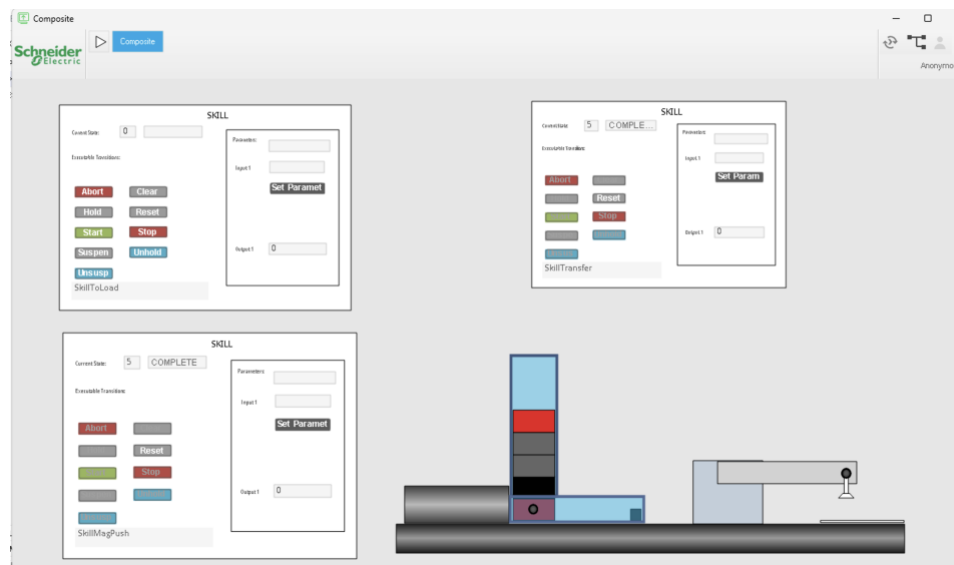


Figure 3. Visualisation of the system and three faceplates of the skills.

You may want to develop ideas and their prototypic implementation towards one or several following research questions:

RQ1. Having a collection of documents describing different facets of an automated production system, establish automatically whether a certain product or process can be produced/ implemented on it. Investigate if SPARQL queries to GraphDB, or LLM based analysis, or combination thereof could help that.

RQ2. How can missing parts (physical or software) be identified and then automatically generated if investigation in RQ1 establishes the plan is not implementable.

RQ3. How can simulation models of the production modules be used in this workflow. Feel free to develop a new simulation model and connect it to the SoftPLC of EcoStruxure tool.

Provided materials:

1. Example of a solution for the FESTO MPS Distribution station implemented in EcoStruxure Automation Expert (v.23.1) using the Skills approach [FESTO_DS_skills_20251117-122832037.sln.zip](#).
2. Example of ontologies describing mechanical, recipe and software of the system in RDF Turtle format (.ttl) and visualisation thereof in .svg: [ontologies](#).
3. Download link to EcoStruxure Automation Expert (v.23.1) – you can use it in demo mode without any license. [v23.1.zip](#).
4. Video, showing the operation of the solution [Video DS with Skills.mp4](#).
5. Companion Powerpoint slide deck [Distributing Station and Skills.pptx](#).